

Qatar-1b

R-2812

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_130626 · created 2026-09-05T01:31:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456469.82708 +/- 6.7e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.015
Transit depth (Rp/Rs)^2	2.05 +/- 0.43 [%]
Orbital Inclination (inc)	84.32 +/- 1.56
Airmass coefficient 1 (a1)	1.336 +/- 0.02
Airmass coefficient 2 (a2)	-0.234 +/- 0.037
Scatter in the residuals of the lightcurve fit is	1.49 %
Best Comparison Star	None
Optimal Aperture	3.3
Optimal Annulus	10.87
Transit Duration (day)	0.0688 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.015 vs 0.1463 (deviation 0.19σ)
Duration fit vs published	0.0688 d vs 0.0692 d (deviation 0.04σ)
Mid-time O–C	-0.1 min
Depth SNR	8.1
Residual scatter	1.49 %

Light curve

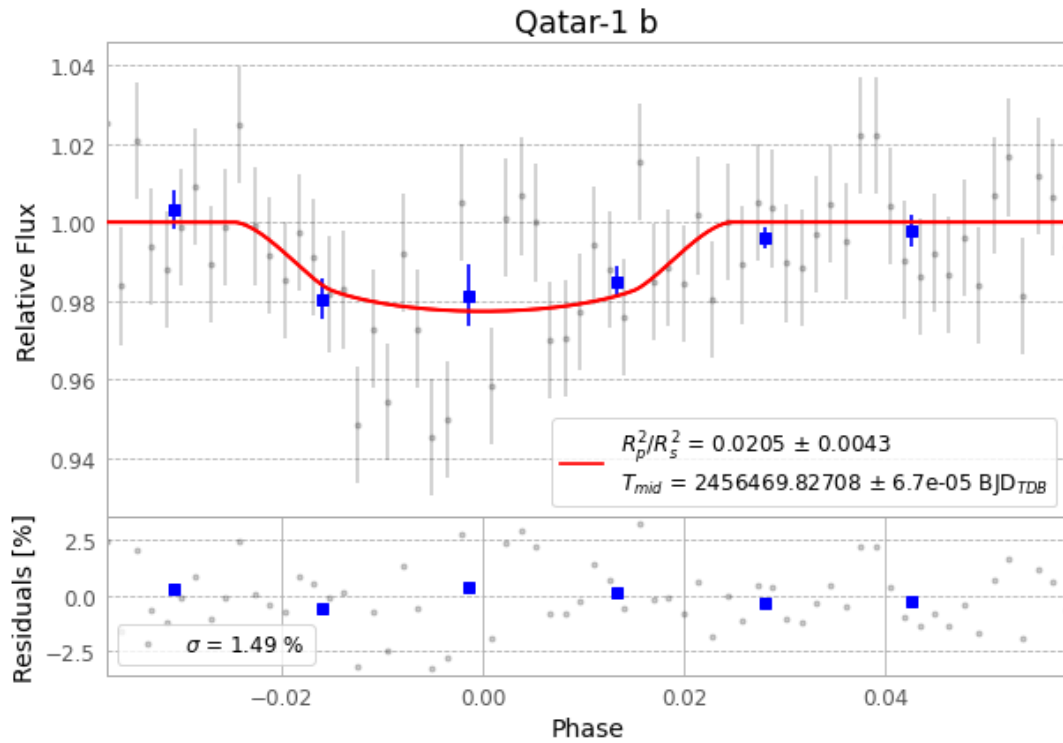


Figure 1 — FinalLightCurve_Qatar-1 b_2013-06-25.png (SHA-256 98e0b26f30b2b71e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [477, 219], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9d2aaa87972f09db...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), Qatar-1130626063312.FITS → Qatar-1130626094812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-130626.log (8aaae34213ad...), FinalLightCurve_Qatar-1 b_2013-06-25.png (98e0b26f30b2...), FinalLightCurve_Qatar-1 b_2013-06-25.csv (3f1e25ff26e5...)

Compute resources

Wall time 395.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 01:31Z.